

# BIG DATA

- What is Big Data?

Big data refers to large, complex sets of data that are difficult to process and analyze using traditional data management tools or methods.

- What is Big Data Analytics?

**Big Data Analytics** refers to the process of examining large and diverse datasets to uncover hidden patterns, correlations, trends, and other useful insights.

The goal is to leverage these insights for decision-making, problem-solving, and predicting future outcomes.

- Case study of Big Data

## 1. Tesla

**Problem:** Tesla collects vast amounts of data from its vehicles, including sensor data, driving behavior, and performance metrics.

**Solution:** Tesla uses big data analytics and machine learning to improve autonomous driving algorithms, predict maintenance needs, and enhance the safety and performance of its vehicles.

## 2. FaceBook

**Problem:** Facebook collects vast amounts of user data to target ads, improve user engagement, and tailor content. It must process and analyze this data in real-time to deliver highly personalized experiences for each user.

**Solution:** Facebook uses sophisticated algorithms and machine learning models to analyze this massive data in real-time to target advertisements, recommend posts, and improve user engagement on the platform.